Faster, Predictable Access to Power

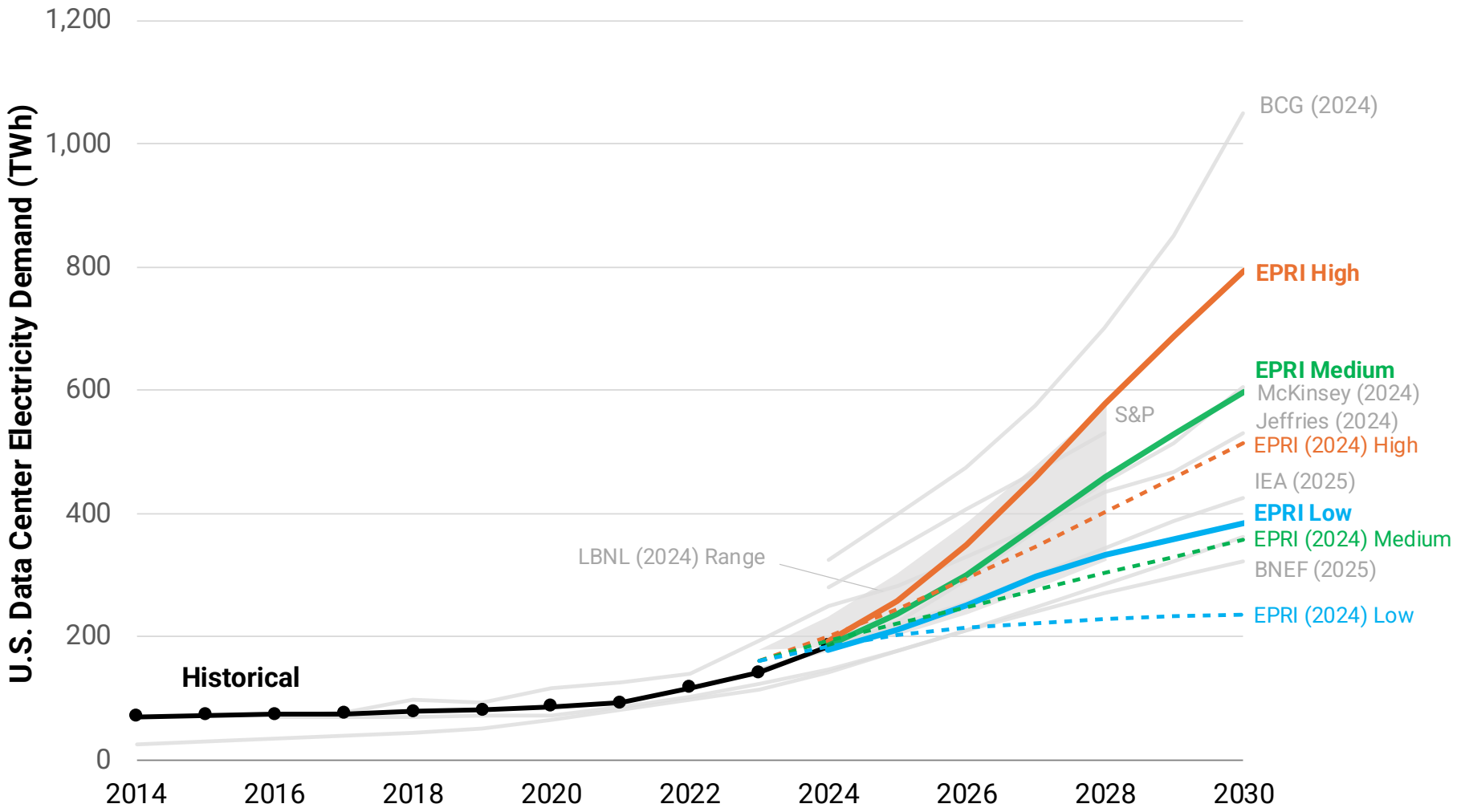
Standardizing AI Data Centers to Unlock Scarce Grid Capacity

Anuja Ratnayake
Emerging Technologies Executive, EPRI

OCP Global Summit 2026
October 13, 2026

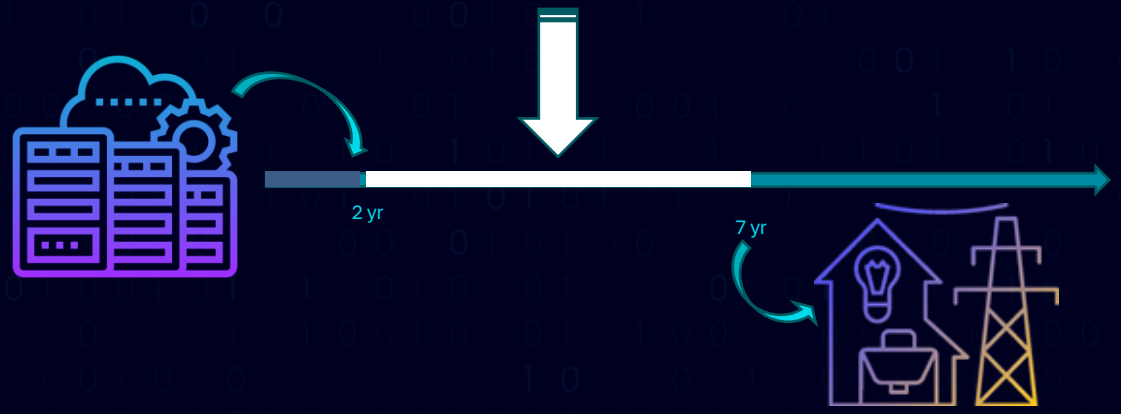
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Data Center Load Growth: Range Across Studies



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Speed to Power Challenge



Solutioning the Gaps



Responsive by Design:
How to be grid responsive without compromising uptime



Integrate with Discipline:
How to integrate flexible load without compromising reliability & affordability

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What Drives Time To Power

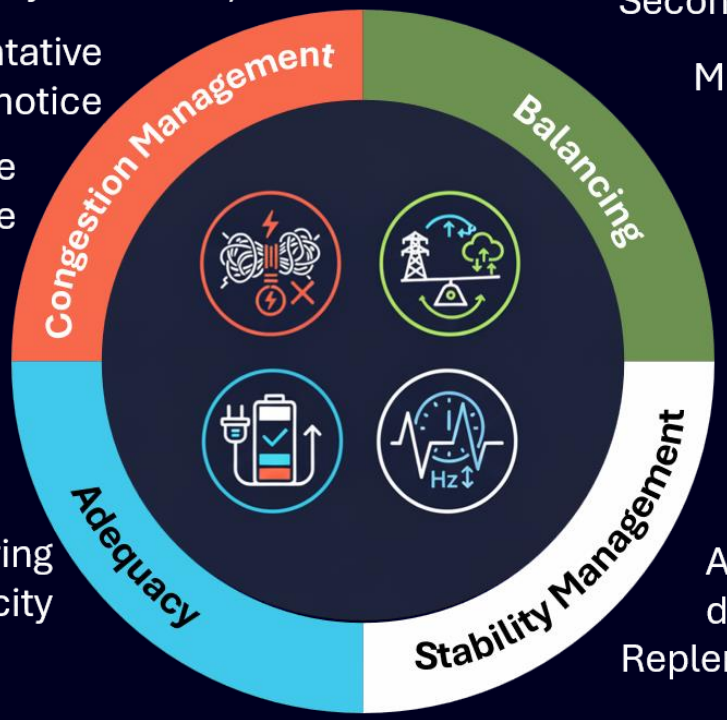
How to think about scarcity in grid capacity

Facilitate planned outages of transmission equipment (days to weeks)

Short duration preventative congestion relief – in day notice

Short duration preventative congestion relief – day ahead notice

Relieve congestion after a fault



Second to second fluctuations

Minute to minute uncertainty

Hour to hour uncertainty

Hour to hour variability

Planned avoidance of system stress

Predictably reduce grid stress during periods of capacity or energy scarcity

Maintain frequency in normal range

Arrest frequency drop after disturbance

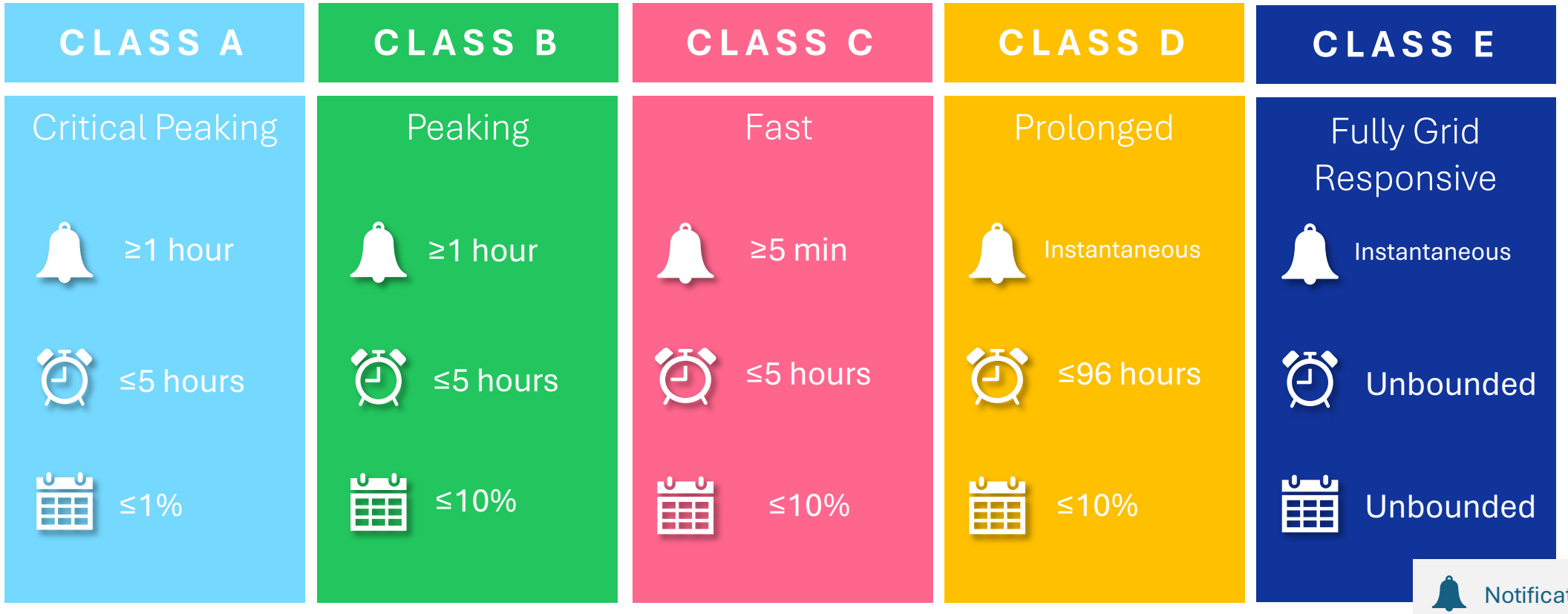
Replenish frequency reserve

Manage Oscillatory loads

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FLEX MOSAIC

INCREASING FLEXIBILITY



Notification Window
Activation Period
Annual Utilization

World's largest cross-sector collaborative on flexibility unveils

Common Flexibility Classification

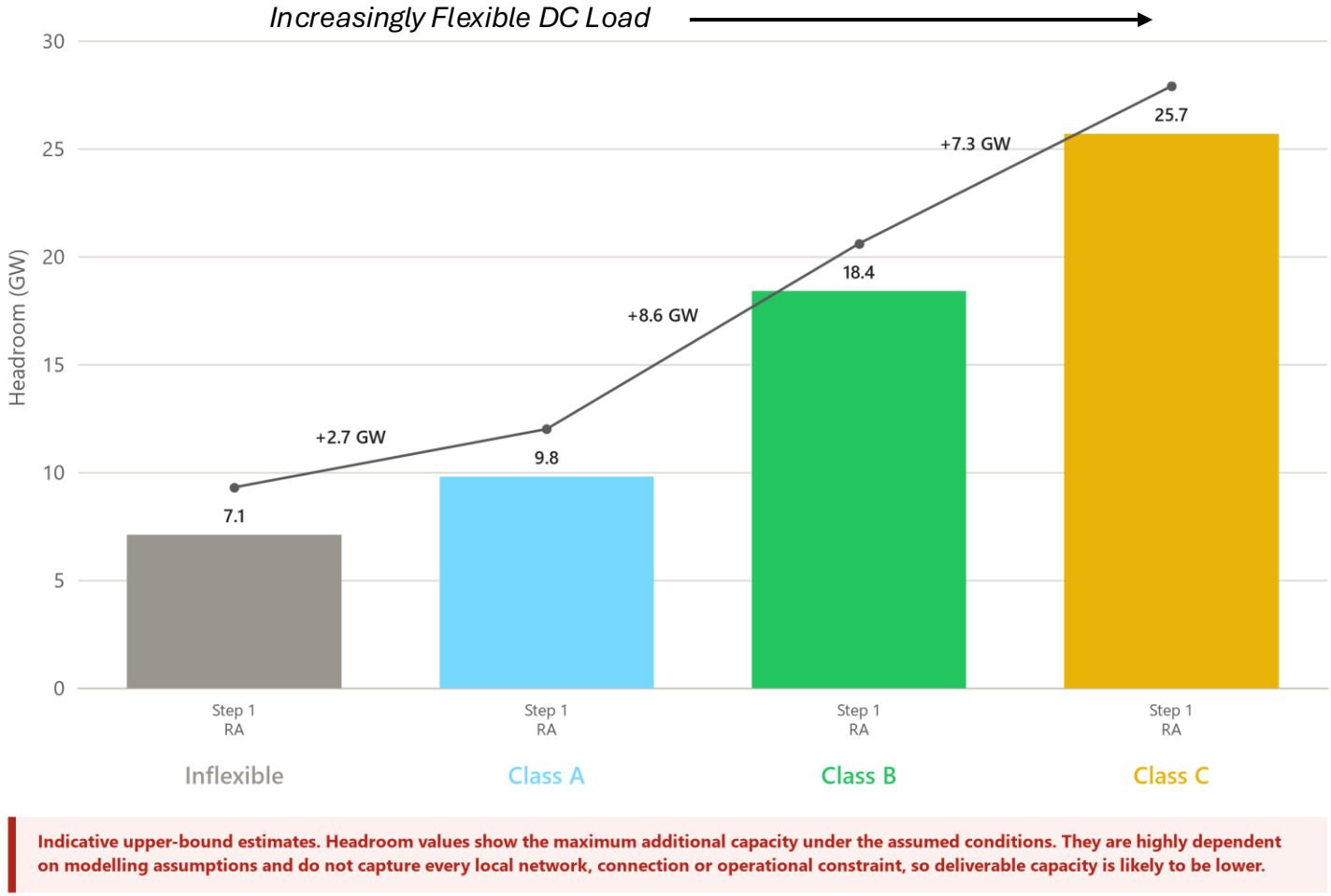
to unlock grid capacity without compromising reliability and affordability

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Case Study:

Flexible operation can make a material difference to amount of additional load that can be added reliably



Indicative upper-bound estimates. Headroom values show the maximum additional capacity under the assumed conditions. They are highly dependent on modelling assumptions and do not capture every local network, connection or operational constraint, so deliverable capacity is likely to be lower.

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NERC CLO Standard

Proposed NERC Registration for total connected load >= 50 MW, supplied at >= 100 kV

COMPUTATIONAL LOAD OWNER | CLO

Entity that owns and maintains the Computational Load Site

DIRECT ACTIONS

- Provide study data, planning models, and model verification documentation
- Maintain applicable disturbance recording and provide requested records
- Establish and use a process for protection system settings

COORDINATION

- Cooperate in interconnection studies and resolve model or data concerns
- Coordinate protection settings across equipment ownership boundaries

Computational Load Owner (CLO) responsible for the **facility, models, protection systems, monitoring infrastructure, and planning data.**

COMPUTATIONAL LOAD OPERATOR | CLOP

Entity that operates the Computational Load Site

DIRECT ACTIONS

- Meet applicable operational data and information specifications
- Comply with Operating Instructions, or promptly report inability to comply
- Maintain communication capability and train personnel on instructions

COORDINATION

- Exchange operational information with grid operating entities
- Coordinate restoration when communication capability fails

Computational Load Operator (CLOP) responsible for the **daily operation of the facility, communications with grid operators, and execution of operating instructions.**

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Join us to unlock speed to power!

https://www.opencompute.org/projects/data-center-facility/energy/

- Become a part of the energy working team
- Contribute with your design

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TOGETHER...SHAPING THE FUTURE OF ENERGY®

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